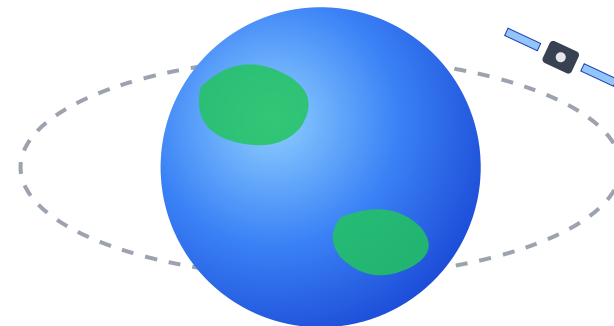


BEFORE YOU BUILD YOUR FIRST MAP

Seeing Earth From Above

An introduction to remote sensing and GIS

Every map you build in GeoLibre starts with a photo taken hundreds of kilometers above your head. Before we get to mapmaking, let's find out how satellites take those photos, and what GIS actually does with them.



What is remote sensing?

Remote sensing means studying or measuring something without touching it. If you look at a bird through binoculars instead of walking up to it, you're already using a simple kind of remote sensing. Scientists do the same thing on a much bigger scale, using cameras and sensors mounted on satellites, airplanes, and drones to study the ground from far away.

Here are a few examples of remote sensing at work:

- A weather satellite watching a storm form over the ocean
- A drone flying over a farm field to check which crops need water
- A camera on the International Space Station photographing a coastline as it passes overhead

WORD OF THE DAY

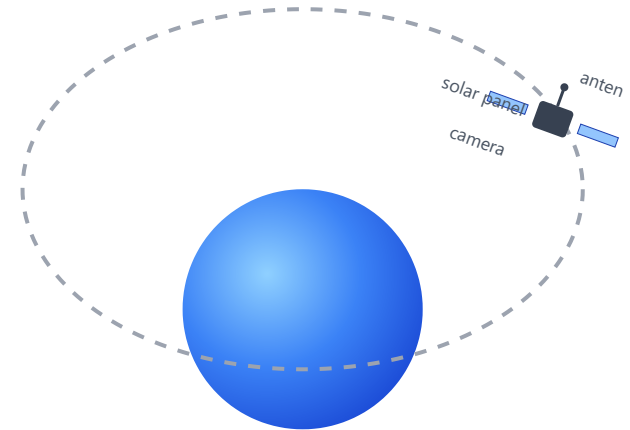
Remote sensing: gathering information about a place or object without making physical contact with it.

What is a satellite?

A satellite is a machine that circles Earth, the way the Moon does, just much closer and usually much smaller. Most Earth-observing satellites carry a few key parts: cameras or sensors to gather images, solar panels to power everything, and an antenna to send those images back down to ground stations on Earth.

FUN FACT

Weather satellites that always watch the same spot on Earth orbit about 36,000 kilometers up, far higher than the satellites that take the sharp, close-up photos used for maps like the one you'll build today.



How do satellites see?

Our eyes can only pick up a narrow slice of the light bouncing off the world around us, the part we call visible light. Many satellites also carry sensors that detect infrared light, which our eyes can't see at all. Healthy plants reflect a lot of infrared light, so scientists can compare infrared images to spot healthy crops and forests versus ones that are dry, stressed, or dying, just by measuring how much infrared light bounces back.

Visible light

What your eyes see: reds, greens, blues, and everything in between, the same colors in a normal photograph.

Infrared light

Invisible to your eyes, but sensors can pick it up. Useful for checking plant health and measuring heat.

TIP

The "Google Satellite" basemap you'll use later in your GeoLibre project is built from photos like these, taken by satellites and aircraft flying over the Earth's surface.

A few more satellite facts

LANDSAT

The first Landsat satellite launched in 1972. Satellites in that same program have photographed Earth's surface every year since, giving scientists an unbroken record of how the planet has changed for more than fifty years.

INTERNATIONAL SPACE STATION

The International Space Station circles Earth about once every 90 minutes. That means the astronauts on board see a sunrise or sunset roughly every 45 minutes.

WORD OF THE DAY

Satellite: a machine that orbits a planet, moon, or star, rather than sitting on the ground.

What is GIS?

GIS stands for Geographic Information System. It's software that takes real information about places, satellite photos, the paths of roads and rivers, the borders of countries and states, and stacks all of it into layers on one map you can study, measure, and draw on.

Think of a GIS map like a sandwich. Each slice is a layer: a satellite photo might be the bottom layer, a country outline goes on top of that, and city names might go on top of everything else. The satellite photo you just learned about is one of those layers. Everything else you draw or add on top of it is GIS at work.

WORD OF THE DAY

GIS: software for storing, organizing, and drawing map layers built from real geographic data.

Explore more on your own

These websites let you browse real satellite images and learn more about space, all built for curious kids.

NASA Space Place

spaceplace.nasa.gov

Games, articles, and activities about space and Earth science, made for kids.

NASA Worldview

worldview.earthdata.nasa.gov

Browse real satellite images of Earth, updated almost every day.

Google Earth

earth.google.com

Spin a 3D satellite globe of the whole planet and fly down to your own neighborhood.

ESA Kids

esa.int/kids

The European Space Agency's kids' portal, with facts about satellites, astronauts, and space missions.

Quick recap

Satellites orbit Earth taking photos using both visible and infrared light. Remote sensing is the general term for studying something from a distance instead of touching it. GIS is the software that takes those photos and other geographic data and stacks them into maps you can study, measure, and draw on, exactly what you're about to do in the next tutorial.

- 1. What does GIS stand for?**
- 2. Name one thing infrared light can help scientists check on the ground.**
- 3. About how often does the International Space Station circle Earth?**

Answers

1. Geographic Information System
2. Plant or crop health (or how dry/stressed plants are)
3. About every 90 minutes